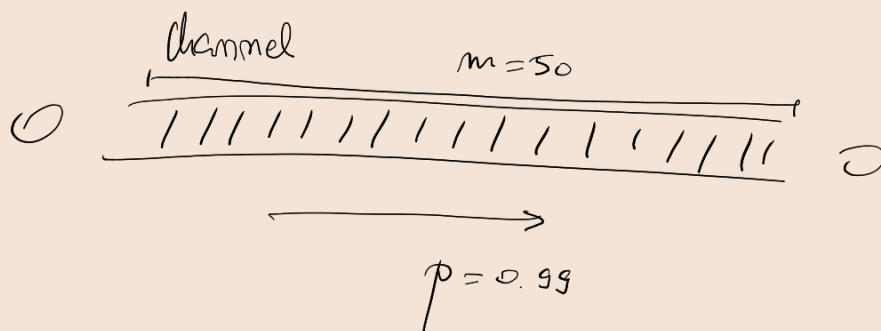




$$C_m^t \cdot p^{m-t} \cdot (1-p)^t$$

$t=0$: Prob. of 0 errors: $C_{50}^0 \cdot p^{50-0} \cdot (1-0.99)^0 = 1 \cdot p^{50} \cdot 1 = p^{50} = (0.99)^{50} = 0.605$

$t=1$: Prob. of 1 error: $C_{50}^1 \cdot 0.99^{50-1} \cdot (1-0.99)^1 = 50 \cdot (0.99)^{49} \cdot 0.01 = 0.3056...$

EAN-13 ← ISBN (books)
← UPC (prod.)

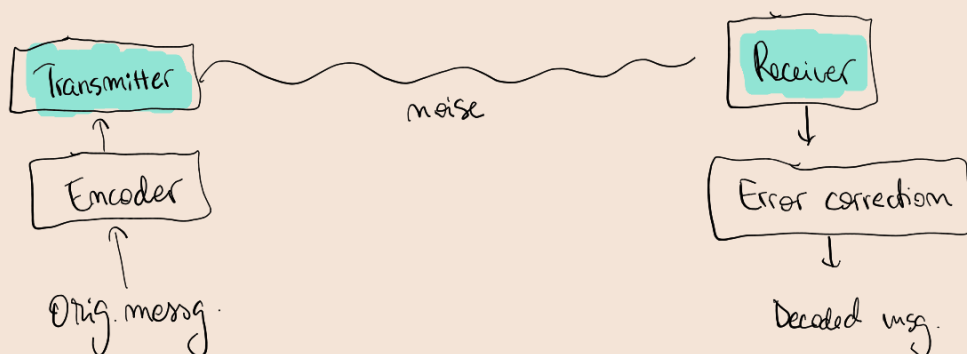
$$a_{13} = \underline{10 - (a_1 + 3a_2 + a_3 + 3a_4 + \dots + a_{11} + 3a_{12}) \% 10}$$

| → 1

|| → 0

Error-correcting codes

CD, DVD, Blue-Ray
 Credit cards
 satellite



Coding problem

↳ BINARY CODES → over finite fields.

SYMM. CHANNELS

$$\begin{array}{l} 1 \xrightarrow{p_1} 0 \\ 0 \xrightarrow{p_2} 1 \end{array} \quad p_1 = p_2 (\%)$$



message of 3 digits

encoder

code word of n digits.

2^3 messages

2^3 code words

2^n possible words.



balance $k, n-k$

Ex: $(3,2)$ - parity check code

Sum modulo 2 = XOR

ENCODE

msg = 00 → code word? → $(0+0) \% 2 = 0 \% 2 = 000$

msg = 01 → $(0+1) \% 2 = 1 \% 2 = 101$

msg = 10 → $(1+0) \% 2 = 1 \% 2 = 110$

msg = 11 → $(1+1) \% 2 = 2 \% 2 = 011$

$\frac{12}{1}$

DECODE

How are they received?

000 → parity? → no (we have no 1's)

101 → parity? → yes (we have an even number (2) of 1's)

110 → yes

011 → yes

111 → no

100 → from where?

Why 101 not here?

101 → yes

111 → no

100 → no

000 → yes

110 → yes

$(3,1)$ - repeating code

ENCODE

msg = 0 → code word: 000

msg = 1 → code word: 111

DECODE

111 → 1

010 → 0

011 → 1

000 → 0

?

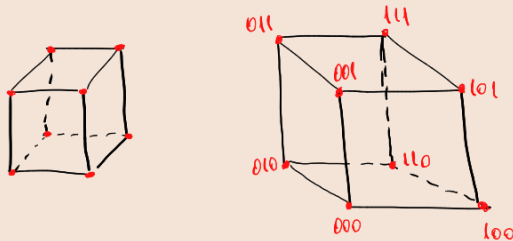
Hamming distance

$$d(u, v) \quad d(101, 100) = 1 \quad \begin{array}{c} 100 \quad 101 \\ 1? \end{array}$$

$$? \left\{ \begin{array}{l} d(110, 001) = 3 \\ \underline{001}, 010, 011, 100, 101, 110 \\ \quad \quad \quad 3 \\ d(101, 011) = 2 \\ \quad \downarrow \\ 011, \underline{100, 101} \\ \quad \quad \quad 2 \end{array} \right.$$

$$\begin{aligned} & \cdot d(u, v) = d(v, u) \\ & \cdot d(u, v) + d(v, w) \geq d(u, w) \\ & \cdot d(u, v) \geq 0 \\ & d(u, v) = 0 \Leftrightarrow \underline{u = v} \\ & (d(u, u) = 0) \end{aligned}$$

$(n-k)$ -code $\Rightarrow 2^m$ received words.



$$\begin{aligned} d(001, 110) &\rightarrow 001 \rightarrow \underline{101} \rightarrow 111 \rightarrow 110 \\ &\rightarrow 001 \rightarrow \underline{011} \rightarrow 111 \rightarrow 110 \\ &\rightarrow 001 \rightarrow \underline{000} \rightarrow 100 \rightarrow 110 \\ &\rightarrow 001 \rightarrow \underline{000} \rightarrow 010 \rightarrow 110 \end{aligned}$$

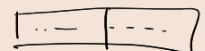
$$\begin{aligned} \sim \text{code } \underline{\text{detects}} \ t (< t) \text{ errors } &\Leftrightarrow \boxed{\min(d(u, v)) \geq t+1} \\ \text{code } \underline{\text{corrects}} \ \dots \dots \dots &\Leftrightarrow \boxed{\min(d(u, v)) \geq 2t+1} \end{aligned}$$

$$a_0 a_1 \dots a_{m-1} \rightsquigarrow \underline{a_0 + a_1 x + \dots + a_{m-1} x^{m-1}} \in \mathbb{F}_2[x] \rightarrow \{0, 1\}$$

$\textcircled{p} \rightarrow$ generator of the code.

$\downarrow$
generates polynomial code = (n, k) -code

$\downarrow$
code words: $\underline{\deg(\text{poly})} \leq m : p$



$$\xrightarrow{k} m \in \mathbb{F}_2[x], \deg(m) < k.$$

$$\textcircled{1} p = 1 + x^2 + x^3 + x^4 \in \mathbb{F}_2[x] \text{ generator poly of a } (7, 3) \text{ code. Encode msg} = 101.$$

$\downarrow$
 $m=7$

$\downarrow$
 $k=3$

$$\text{msg } 101 \rightarrow m = 1 \cdot 1 + 0 \cdot x + 1 \cdot x^2 = \underline{1 + x^2}$$

$$\rightarrow m \cdot x^{n-k} = (1 + x^2) \cdot x^{7-3} = (1 + x^2) \cdot x^4 = \underline{x^4 + x^6}$$

$$\rightarrow r = \underline{m \cdot x^{n-k}} \% p = (x^4 + x^6) \% (1 + x^2 + x^3 + x^4) = \underline{1 + x}$$

$$\begin{array}{r} x^6 + x^4 \\ - x^6 - x^5 - x^4 - x^2 \\ \hline -x^5 - x^2 \\ x^5 + x^4 + x^3 + x \end{array} \quad \begin{array}{r} x^4 + x^3 + x^2 + 1 \\ x^2 - x - 1 + 1 \end{array}$$

$$\begin{array}{r} x^4 + x^3 - x^2 + x \\ -x^4 - x^3 - x^2 - 1 \\ \hline -2x^2 + x - 1 \end{array} \quad ?$$

$$\Rightarrow r = 1 + x + (x^4 + x^6) = 1 + x + x^4 + x^6$$

$\Rightarrow$ code word: (1100101)

② $p = 1 + x + x^3 \in \mathbb{F}_2[x]$ gen. pol. of a (6-3) code.

100011 } detectable errors?
100110

100011 $\rightarrow$ code word? $\rightarrow$ poly: p ?
100110 $\rightarrow$

③ All the code words of (6-3)-code gen. by $p = 1 + x + x^3 \in \mathbb{F}_2[x]$.

$n=6$
 $k=3$ $\rightarrow 2^k = 8$ code words.

$000 \rightarrow 000000$
 $001 \rightarrow 111001$
 $010 \rightarrow 011010$
 $011 \rightarrow 100011$
 $100 \rightarrow 010100$
 $101 \rightarrow 101101$
 $110 \rightarrow 001110$
 $111 \rightarrow 110111$

how are they obtained?

$$110 \rightarrow m = 1 + x$$

$$\rightarrow m \cdot x^{n-k} = (1+x) \cdot x^3 = x^3 + x^4$$

$$\rightarrow r = m \cdot x^{n-k} \% p = (x^3 + x^4) \% (1 + x + x^3) = 1 + x^2$$

$$\rightarrow r = r + m \cdot x^{n-k} = 1 + x^2 + x^3 + x^4$$

$$\begin{array}{r|l} x^4 + x^3 & x^3 + x + 1 \\ -x^4 - x^3 - x & x \\ \hline -x & \end{array}$$

??

Seminar 12

(3,2)-parity check code

2 digits message

3 digits code

first digit = (sum of the 2 digits in the message) % 2

- ① a) $110 \rightarrow 1 = (1+0) \% 2 = 1 \% 2 = 1 \Rightarrow$ no detectable errors.
 $010 \rightarrow 0 = (1+0) \% 2 = 1 \% 2 = 1 \Rightarrow$ detectable error.
 $001 \rightarrow 0 = (0+1) \% 2 = 1 \% 2 = 1 \Rightarrow 1 \text{ --- } 1 \text{ ---}$
 $111 \rightarrow 1 = 2 \% 2 = 0 \Rightarrow 1 \text{ --- } 1 \text{ ---}$
 $101 \rightarrow 1 = 1 \% 2 = 1 \Rightarrow$ no det. errors
 $000 \rightarrow$ no det. errors.

Which digit is the message?

- b) $(3,1)$ - repeating code
 1 digit message
 3 digits code
 first, second digits repeat the code.

So the message is the last digit?

- $111 \rightarrow 11$ repeat the message
 $011 \rightarrow 01$ repeat the message
 $101 \rightarrow 10$ repeat " --- "
 $010 \rightarrow 01$ repeat ?
 $000 \rightarrow 00$ repeat the message 0?
 $001 \rightarrow 00$ does not repeat the message

- ② $f = x^7 + x^6 + x^4 + x^3 + 1$
 $g = x^6 + x^3 + x^2 + x$ } code words in the poly $(8,4)$ -code generated by $p = 1 + x^2 + x^3 + x^4 \in \mathbb{Z}_2[x]$?

$m=8$
 $k=4$

$$\begin{array}{r|l} f & p \\ x^7 + x^6 + x^4 + x^3 + 1 & x^4 + x^3 + x^2 + 1 \\ -x^7 - x^6 - x^5 - x^3 & x^3 + x \\ \hline -x^5 + x^4 + 1 & \\ x^5 + x^4 + x^3 + x & \end{array}$$

$x^3 + x + 1 \neq 0 \Rightarrow p \nmid f \Rightarrow f$ not a code word.

g p

$$\begin{array}{r|l} x^6 + x^3 + x^2 + x & x^4 + x^3 + x^2 + 1 \\ -x^6 - x^5 - x^4 - x^2 & x^2 - x \\ \hline -x^5 - x^4 + x^3 + x & \\ x^5 + x^4 + x^3 + x & \\ \hline 2x^3 + 2x = 0 & \Rightarrow g \text{ is a code word.} \\ \mathbb{Z}_2! & \end{array}$$

- ③ $(6,3)$ -code $p = 1 + x^2 + x^3 \in \mathbb{Z}_2[x]$
 $m=6$
 $k=3$
- $\sum_{i=0}^2 m_i x^i$ $\rightarrow$ $m = 000 = 0 \cdot x^0 + 0 \cdot x^1 + 0 \cdot x^2 = 0$
 $m \cdot x^{m-k} = 0$
 $0 = m \cdot x^{m-k} \pmod{p} = 0$

$$2^k = 2^3 = 8 \text{ words.}$$

$$\begin{matrix} 010 \\ 011 \\ 100 \\ 101 \\ 110 \\ 111 \end{matrix}$$

$$r = h + m \cdot x^{n-k} = 0.$$

$$\Rightarrow \frac{000000}{n \text{ digits}}$$

[001]

$$m = 001 = 0 \cdot x^0 + 0 \cdot x^1 + 1 \cdot x^2 = x^2.$$

$$m \cdot x^{n-k} = x^2 \cdot x^{6-3} = x^2 \cdot x^3 = x^5.$$

$$h = m \cdot x^{n-k} \% p = x^5 \% (x^3 + x^2 + 1) = x + 1, \quad r = h + m \cdot x^{n-k} = x + 1 + x^5 = \underline{1 \ 1 \ 0 \ 0 \ 0 \ 1}$$

$$\begin{array}{r} x^5 \overline{) x^3 + x^2 + 1} \\ -x^5 - x^2 - x \\ \hline -x^4 - x^2 \\ x^4 + x^3 + x \\ \hline x^3 - x^2 + x \\ -x^3 - x^2 - 1 \\ \hline -2x^2 + x - 1 = x + 1 \\ \hline 0 \end{array}$$

$-1 \equiv 1 \pmod{2}$

[010]

$$m = 010 = 0 \cdot x^0 + 1 \cdot x^1 + 0 \cdot x^2 = x.$$

$$m \cdot x^{n-k} = x \cdot x^3 = x^4.$$

$$h = m \cdot x^{n-k} \% p = x^4 \% (x^3 + x^2 + 1) = x^2 + x + 1.$$

$$r = h + m \cdot x^{n-k} = x^4 + x^2 + x + 1.$$

$$\begin{array}{r} x^4 \overline{) x^3 + x^2 + 1} \\ -x^4 - x^3 - x \\ \hline 1 - x^3 - x \\ x^3 + x^2 + 1 \\ \hline x^2 - x + 1 = x^2 + x + 1 \end{array}$$

$$\underline{1 \ 1 \ 1 \ 0 \ 1 \ 0}$$

[011]

$$m = 011 = 0 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 = x + x^2.$$

$$m \cdot x^{n-k} = (x + x^2) \cdot x^3 = x^4 + x^5.$$

?

$$h = (m \cdot x^{n-k}) \% p = (x^5 + x^4) \% (x^3 + x^2 + 1) = x^2$$

$$\begin{array}{r} x^5 + x^4 \overline{) x^3 + x^2 + 1} \\ -x^5 - x^4 - x^2 \\ \hline -x^2 = x^2 \end{array}$$

$$x^2(x^3 + x^2 + 1) = x^5 + x^4 + x^2.$$

[100]

$$m = 1 \cdot x^0 + 0 \cdot x^1 + 0 \cdot x^2 = 1.$$

$$m \cdot x^{n-k} = 1 \cdot x^3 = x^3.$$

$$h = x^3 \% (x^3 + x^2 + 1) = x^2 + 1.$$

$$r = h + m \cdot x^{n-k} = x^3 + x^2 + 1.$$

$$\begin{array}{r} x^3 \overline{) x^3 + x^2 + 1} \\ -x^3 - x^2 - 1 \\ \hline -x^2 - 1 \\ \hline = x^2 + 1 \end{array}$$

$$\underline{1 \ 0 \ 1 \ 1 \ 0 \ 0}$$

[101]

$$m = 1 \cdot x^0 + 0 \cdot x^1 + 1 \cdot x^2 = x^2 + 1.$$

$$m \cdot x^3 = (x^2 + 1) \cdot x^3 = x^5 + x^3.$$

$$h = (x^5 + x^3) \% (x^3 + x^2 + 1) = x^2 + x, \quad r = x^5 + x^3 + x^2 + x.$$

$$\begin{array}{r}
 x^5 + x^3 \quad | \quad x^3 + x^2 + 1 \\
 -x^5 - x^4 - x^2 \quad | \quad x^2 - x + 2 = x^2 + x \\
 \hline
 -x^4 + x^3 - x^2 \\
 x^4 + x^3 + x \\
 \hline
 2x^3 - x^2 + x \\
 -2x^3 - 2x^2 - 2 \\
 \hline
 -3x^2 + x - 2 = -x^2 + x = x^2 + x
 \end{array}$$

$$0 \ 1 \ 1 \ 1 \ 0 \ 1$$

④ $G = \left(\frac{P}{I_3} \right) \in \mathcal{M}_{5,3}(\mathbb{Z}_2)$ | Parity Check matrix?
Code words?

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

$$n=5 \rightarrow m-k=5-3=2$$

$$H = (I_{m-k} | P) = (\overset{\text{red}}{I_2} | \overset{\text{green}}{P}) = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$u = (u_1, u_2, u_3, u_4, u_5) \text{ code vector if } H \cdot u = 0_2$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \end{pmatrix}_{2 \times 5} \cdot \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{pmatrix}_{5 \times 1} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}_{2 \times 1}$$

$$\Rightarrow \begin{pmatrix} u_1 & 0 & 0 & 0 & u_5 \\ 0 & u_2 & u_3 & u_4 & u_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} u_1 + u_5 = 0 \Rightarrow \underline{u_5 = -u_1} \\ u_2 + u_3 + u_4 + u_5 = 0 \Rightarrow u_2 + u_3 + u_4 - u_1 = 0 \\ \Rightarrow \underline{u_1 = u_2 + u_3 + u_4} \end{cases}$$

$$\Rightarrow u = (u_1, u_2, u_3, u_4, u_5) = (u_2 + u_3 + u_4, u_2, u_3, u_4, \underline{-u_2 - u_3 - u_4})$$

$$\Rightarrow 2^{K=3} \text{ code words. } (8).$$

$$\Rightarrow \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 1 \\ 1 \\ -1 \end{pmatrix} \right\}$$

Recall

(3,2)-parity check code

- 2 digits message
- 3 digits code (encode)
- 1st digit = sum of the 2 digits in the message

(3,1)-repeating code

- 1 digit message
- 3 digits code
- 1st, 2nd digits repeat the code

Parity-check matrix

$$H = (I_{m-k} | P)$$

$$u \in \mathcal{M}_{m,1}(\mathbb{Z}_2) \text{ is a code vector}$$

$$\Leftrightarrow H \cdot u = 0$$

(mod 2)

Generator of a polynomial code

$p \in \mathbb{Z}_2[x]$ of degree $m-k$

↓
generator of a polynomial code (n, k) ,
whose words are polynomials of degree less than n , $\therefore p$.

↓

for a $(n-k)$ polynomial code $\rightarrow 2^k$ code words.

for a message m : $m \cdot x^{n-k} = 2p + r$ degree(r) < degree(p) = $n-k$.

$\rightarrow$ we code it as: $r = r + m \cdot x^{n-k}$.

Hamming distance

u, v of the same length $\Rightarrow$ the nr. of positions in which they differ. $\rightarrow d(u, v) \rightarrow$ metric on $\mathbb{Z}_2^n$.

A code detects all errors $\leq t$:

$$\min(d(u, v)) \geq t+1$$

A code corrects all errors $\leq t$:

$$\min(d(u, v)) \geq 2t+1$$

Encoder

$$\mathcal{E}: \mathbb{Z}_2^k \rightarrow \mathbb{Z}_2^n \text{ with } [\mathcal{E}]_{EE'} = G.$$

⑤ $G = \begin{pmatrix} P \\ I_n \end{pmatrix} \in \mathcal{M}_{9,4}(\mathbb{Z}_2) \rightarrow 2^4 = 16$ code words?

$$H = (I_5 | P) \Rightarrow H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

For $u = (u_1, \dots, u_8)$ we compute $H \cdot u = 0_9$

$$\Rightarrow \begin{cases} u_1 + u_8 = 0 \Rightarrow u_1 = -u_8 \Rightarrow \underline{u_1 = u_8} \\ u_2 + u_7 + u_9 = 0 \Rightarrow u_2 = -u_7 - u_9 = \underline{u_7 + u_9} \\ u_3 + u_6 + u_8 + u_9 = 0 \Rightarrow u_3 = -u_6 - u_8 - u_9 = \underline{u_6 + u_8 + u_9} \\ u_4 + u_7 = 0 \Rightarrow u_4 = -u_7 = \underline{u_7} \\ u_5 + u_6 + u_9 = 0 \Rightarrow u_5 = -u_6 - u_9 = \underline{u_6 + u_9} \end{cases}$$

$$\Rightarrow u = (u_8, u_7 + u_9, u_6 + u_8 + u_9, u_7, \underline{u_6 + u_9}, u_6, u_7, u_8, u_9)$$

$$\min(d(u_i, u_j))$$

$$\min(d(u_1, u_i)) = \min(d(u_2, u_i)) = \dots = \min(d(u_9, u_i)) = 3.$$

$$\min(d(u_i, u_j)) = 3 \geq t+1$$

$$\Rightarrow t \leq 2 \Rightarrow \text{code detects 2 errors.}$$

$$\min(d(u_i, u_j)) = 3 \geq 2t+1$$

$$\Rightarrow t \leq 1 \Rightarrow \text{code corrects 1 error.}$$

⑥ Encode $\begin{pmatrix} 1101 \\ 0111 \\ 1000 \end{pmatrix}$ using the generator matrix of the $(9,4)$ -code of Ex 5.

$$\begin{pmatrix} 0000 \\ 1000 \end{pmatrix}$$

$$[\mathcal{S}]_{EE'} = G \Rightarrow \begin{cases} \mathcal{S}(e_1) = \mathcal{S}(1000) \dots 001011000 \\ \mathcal{S}(e_2) = \mathcal{S}(0100) \dots 010100100 \\ \mathcal{S}(e_3) = \mathcal{S}(0010) \dots 101000010 \\ \mathcal{S}(e_4) = \mathcal{S}(0001) \dots 011010001 \end{cases}$$

$$1101 = e_1 + e_2 + e_4 \Rightarrow \mathcal{S}(1101) = \mathcal{S}(e_1 + e_2 + e_4) \stackrel{\text{why}}{=} \mathcal{S}(e_1) + \mathcal{S}(e_2) + \mathcal{S}(e_4) \\ = 001011000 + 010100100 + 011010001 \\ = 000101101.$$

$$0111 = e_2 + e_3 + e_4 \Rightarrow \mathcal{S}(0111) = \mathcal{S}(e_2) + \mathcal{S}(e_3) + \mathcal{S}(e_4) = 100110111.$$

$$0000 = e_1 + e_2 \Rightarrow \dots$$

$$1000 = e_1 \Rightarrow \dots$$

⑦ Generator matrix =? Parity check matrix?

$(4,1)$ -code gen. by $p = 1 + x + x^2 + x^3 \in \mathbb{Z}_2[x]$.

$\mathcal{S}: \mathbb{Z}_2^1 \rightarrow \mathbb{Z}_2^4$ $\left\{ \begin{array}{l} \text{Every codeword has } m \text{ bits (4).} \\ k=1 \rightarrow \text{choose 1 message bit (K)} \end{array} \right.$

$$[\mathcal{S}]_{EE'} = G$$

$$E = (e_1) = 1$$

$$E' = (e'_1, e'_2, e'_3, e'_4)$$

a 4-bit word (c_0, c_1, c_2, c_3)

$\downarrow$

$$c_0 + c_1x + c_2x^2 + c_3x^3.$$

$p = 1 + x + x^2 + x^3$ corresponds to 1111 = message

$$e_1 = 1 \Rightarrow m = 1 \Rightarrow m \cdot x^{n-k} = 1 \cdot x^3 = x^3$$

$$r = (m \cdot x^{n-k}) \% p = x^3 \% (1 + x + x^2 + x^3) = x^2 + x + 1.$$

$$r = x^3 + x^2 + x + 1 \Rightarrow 1111.$$

$$\begin{array}{r} x^3 \mid x^3 + x^2 + x + 1 \\ -x^3 - x^2 - x - 1 \\ \hline x^2 + x + 1 \end{array}$$

$$\Rightarrow G = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} P \\ I_k \end{bmatrix} \Rightarrow P = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$H = (\underbrace{I_{n-k}}_{I_3} \mid P) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

⑧ Generator matrix =? Parity check matrix?

$(7,3)$ -code generated by $p = 1 + x^2 + x^3 + x^4 \in \mathbb{Z}_2[x]$.

$$\dots (3) \dots (7)$$

$$\mathbb{F}_2: \mathbb{F}_2 \rightarrow \mathbb{F}_2$$

$$[g]_{EE'} = G$$

$$E = (e_1, e_2, e_3)$$

$$E' = (e_1', e_2', e_3', e_4', e_5', e_6', e_7')$$

$$e_1 = (1, 0, 0) \Rightarrow 100 = 1 \cdot x^0 + 0 \cdot x^1 + 0 \cdot x^2 = 1 = m$$

$$m \cdot x^{n-k} = 1 \cdot x^4 = x^4$$

$$r = x^4 \% (x^4 + x^3 + x^2 + 1) = x^3 + x^2 + 1$$

$$\begin{array}{r|l} x^4 & x^4 + x^3 + x^2 + 1 \\ -x^4 - x^3 - x^2 - 1 & 1 \\ \hline & 1 \end{array}$$

$$r = x^4 + x^3 + x^2 + 1$$

$$\Rightarrow \begin{array}{cccccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 \end{array}$$

$$e_2 = (0, 1, 0) \Rightarrow 010 = 0 \cdot x^0 + 1 \cdot x^1 + 0 \cdot x^2 = x$$

$$m \cdot x^{n-k} = x \cdot x^4 = x^5$$

$$r = x^5 \% (x^4 + x^3 + x^2 + 1) = x^2 + x + 1$$

$$\begin{array}{r|l} x^5 & x^4 + x^3 + x^2 + 1 \\ -x^5 - x^4 - x^3 - x & x - 1 \\ \hline -x^4 - x^3 - x & \\ x^4 + x^3 + x^2 + 1 & \\ \hline x^2 - x + 1 & \\ x^2 + x + 1 & \end{array}$$

$$r = 1 + x + x^2 + x^5$$

$$\begin{array}{cccccc} 1 & 1 & 1 & 0 & 0 & 1 & 0 \end{array}$$

$$e_3 = (0, 0, 1) \Rightarrow 001 = 0 \cdot x^0 + 0 \cdot x^1 + 1 \cdot x^2 = x^2 = m$$

$$m \cdot x^{n-k} = x^2 \cdot x^4 = x^6$$

$$r = x^6 \% (x^4 + x^3 + x^2 + 1) = x^3 + x^2 + x$$

$$\begin{array}{r|l} x^6 & x^4 + x^3 + x^2 + 1 \\ -x^6 - x^5 - x^4 - x^2 & x^2 - x \\ \hline -x^5 - x^4 - x^2 & \\ x^5 + x^4 + x^3 + x & \end{array}$$

$$r = x^6 + x^3 + x^2 + x$$

$$\begin{array}{cccccc} 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{array}$$

$$\begin{array}{r} / x^3 - x^2 + x \\ \hline = x^3 + x^2 + x \end{array}$$

$$G = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ \hline 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad P \text{ (parity check matrix)}$$

$$I_3 (I_K)$$

$$\Rightarrow P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

parity check matrix

$$\Rightarrow H = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$\uparrow$
 I_{m-k}
 $\uparrow$
 I_k

$\uparrow$
 P

